network *sampling*

introduction to *network analysis* (*ina*)

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sampling *overview*

— *snowball sampling* and *contact tracing* in *sociology*

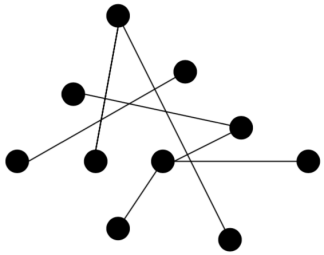
network-based methods for sampling hidden populations

— *fractality* and *self-similarity* of *real networks* [SHM05]

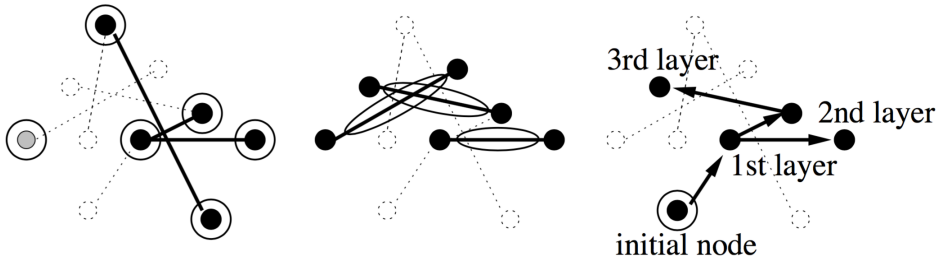
fractality is property of object that it is similar to part of itself

self-similarity demands power-law size scaling under renormalization

— any *real network* is just *sample* of *true network* [BŠWB15]



original network



node/link selection and snowball sampling

sampling *populations*

— *snowball sampling* similar to *breadth-first search*

nodes sampled proportional to their eigenvector centrality

— *contact tracing* similar to biased *best-first search*

nodes sampled according to some hidden variable

— *respondent-driven sampling* similar to *random walk*

nodes sampled proportional to their degree

estimate $\langle x \rangle = \frac{\sum_i x_i}{\sum_i 1}$ corrected $\langle x \rangle = \frac{\sum_i x_i / k_i}{\sum_i 1 / k_i}$

sampling *methods*

— *random selection methods* for *global network sparsification*

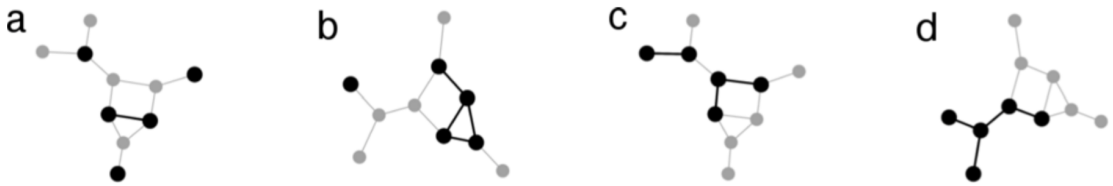
random node/link selection w/ or w/o induction etc.

— *network exploration methods* for *local network sampling*

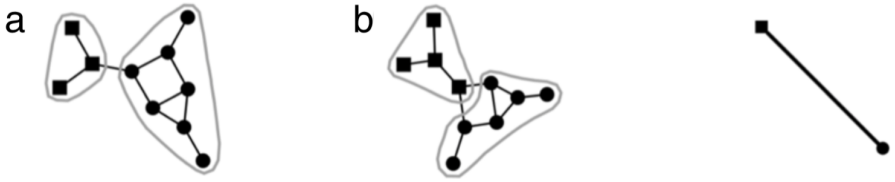
random walks, snowball and expansion sampling etc.

— *merging/aggregation methods* for *network simplification*

box covering, cluster growing, community aggregation etc.



(a) node selection (b) by degree (c) link selection and (d) snowball sampling



(a) community aggregation and (b) cluster growing

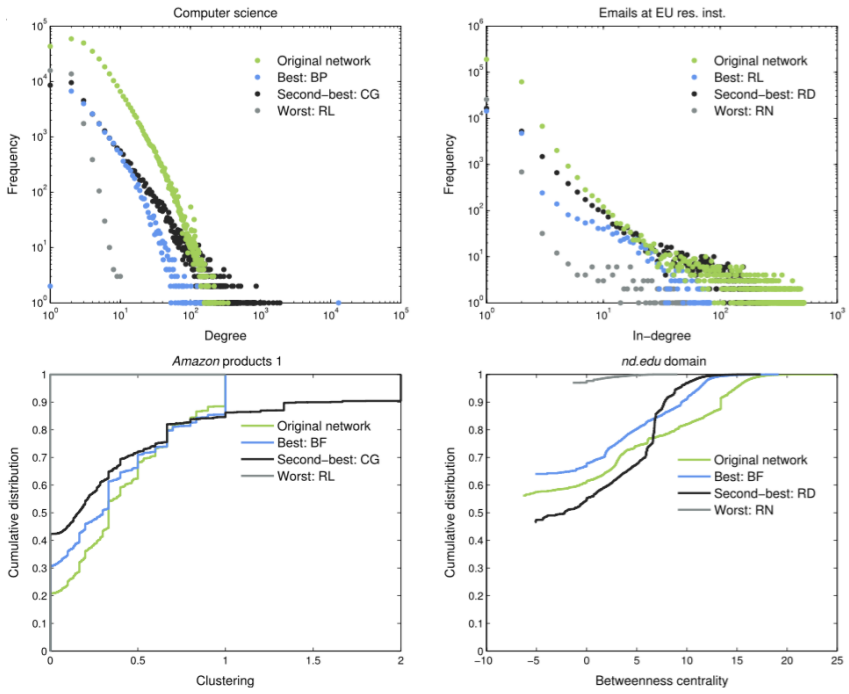
sampling *networks*

— *similar sampled networks* for $\gtrsim$ 15% *network* [LF06]

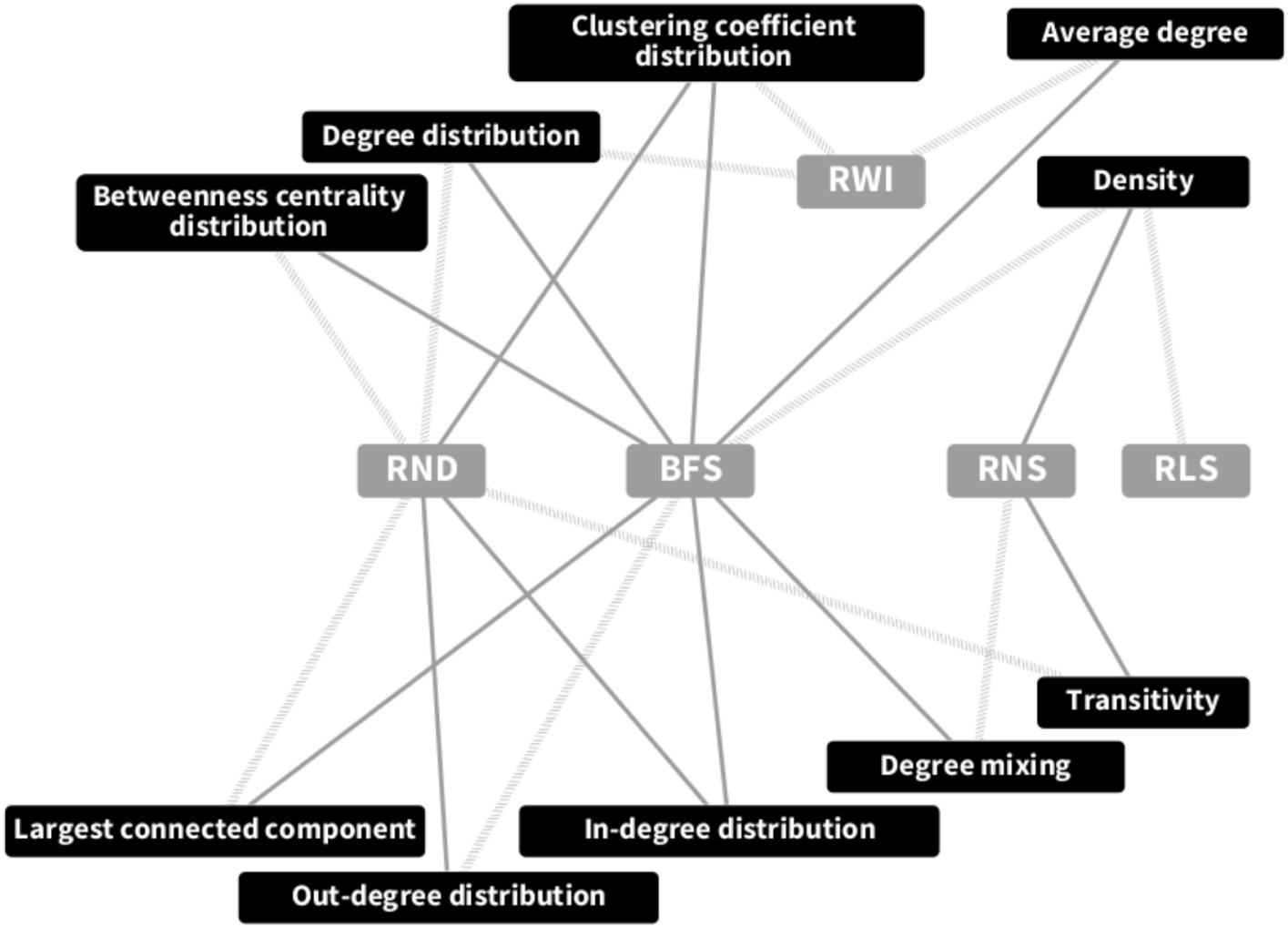
— *methods indistinguishable* for $\ll$ 1% *network* [BŠB17]

— *selection/exploration* $\ggg$ *aggregation methods* [BŠB14]

independent of network type or size but not sample size



sampling *scheme*



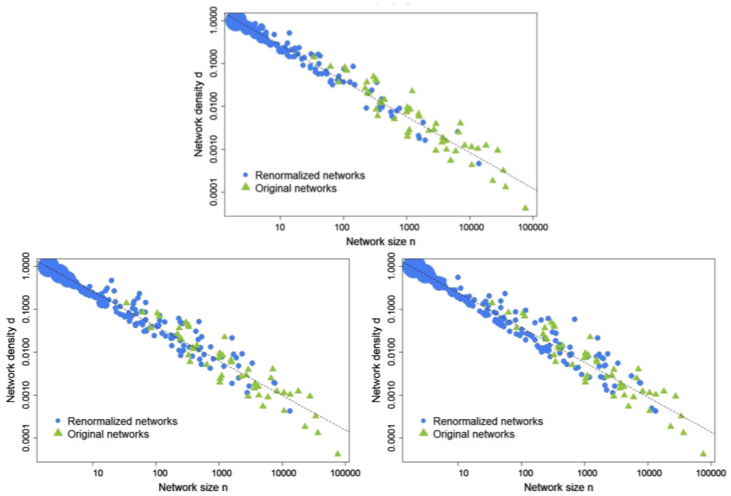
sampling *density*

— *self-similar network density* of *real networks* [LJT+11]

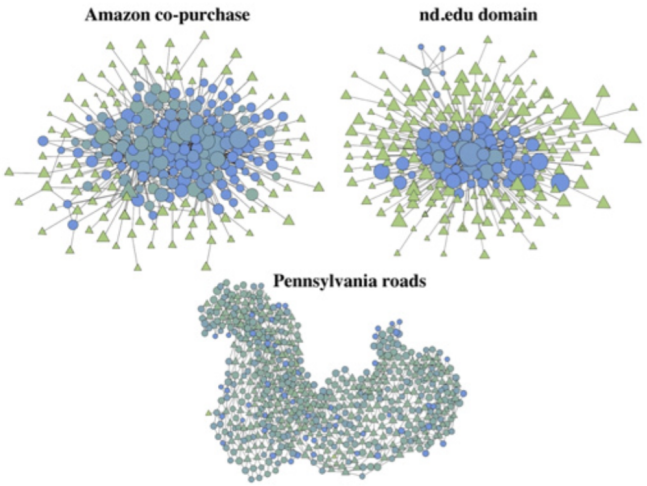
most real networks are sparse with $\rho \approx 7.89n^{-0.99}$

— *network density invariant* under *any renormalization* [BŠB12]

box covering, cluster growing, community aggregation etc.



density invariance under renormalization



renormalized networks

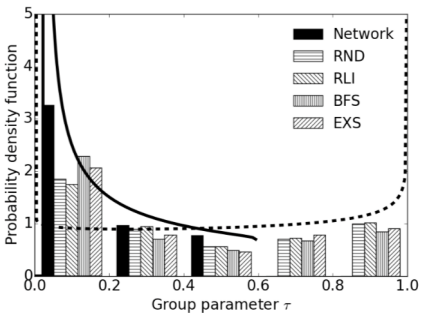
sampling *communities*

— *pronounced community structure* in *real networks* [GN02]

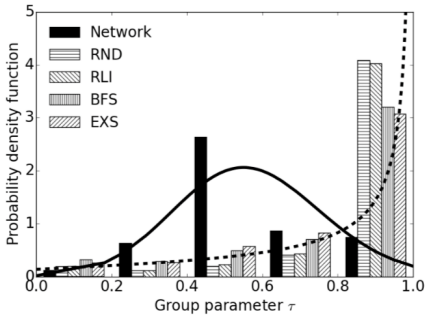
many real networks contain communities with $\tau \gg 0$

— *community structure enhanced* under *any sampling* [BŠWB15]

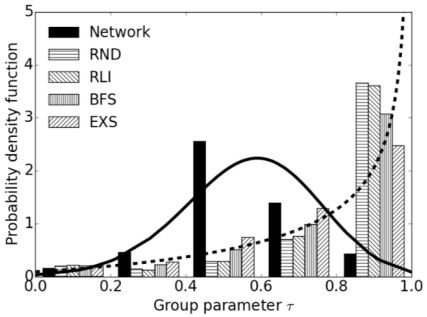
random selection, snowball and expansion sampling etc.



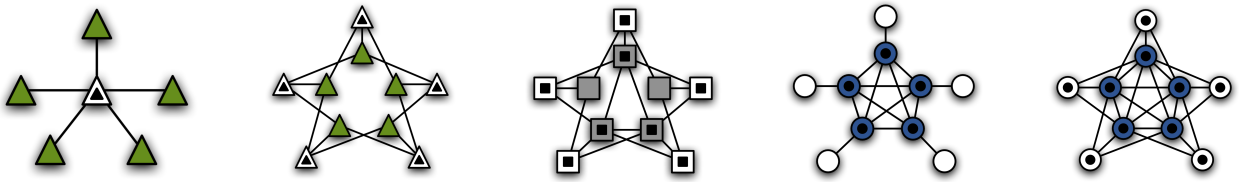
citation network



social network



collaboration network



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